Data Flow Diagram (DFD)

Template: Data Flow Diagram & Symbol Legend	Maximum Marks: 2 Marks
Date: 15 March 2026	**Team ID:** FV-AI-2026-X1
Project Name: FoodVerse AI	**Track:** AI/ML & Generative AI

DFD Symbol Legend

Create a Data Flow Diagram (DFD) to show how data moves through your system — between external entities, processes, and data stores. Use the symbol legend below as a guide.

Symbol	Symbol Name	Description & FoodVerse AI Application
Oval / Rounded	External Entity	User / Cook / Health Auditor initiating food scans, voice notes, or biometric goal configurations.
Numbered Rect	Process	Automated computational transform (1.0 Multimodal Ingestion, 2.0 CV Spoilage Analysis, 3.0 Hygiene Frame Sampling, 4.0 Chef LLM Reasoning, 5.0 Metabolic Plan Generation).
Solid Rect	Data Store	Temporary buffer storage (D1 Temp Uploads), Local Nutrition Cache (D2 Macro Heuristics), Persistent User Profile Store (D3 Profiles).
Labeled Arrow	Data Flow	Movement of data packets: Image bytes, audio WAV, JSON payloads, macro analytics, hygiene compliance verdicts.

Level 0 & Level 1 System Data Flow Narrative

DFD Level	Source / Entity	Process Executed	Target Data Store / API	Output Generated
Level 0 (Context)	End User	0.0 FoodVerse AI Platform Unified Interface	Cloud Groq Inference API + Local Backend	Nutritional facts, safety grade, cooking steps, meal plan
Level 1 (Sub-Process 1.0)	Camera / Mic	1.0 Ingest & Transcribe Media	D1: Temp Upload Directory	Standardized RGB JPEG + Transcribed Context String
Level 1 (Sub-Process 2.0)	Produce Photo	2.0 Extract CV Features & Color Consistency	OpenCV Laplacian Engine	Discoloration ratio, roughness score, anomaly descriptors
Level 1 (Sub-Process 3.0)	Kitchen Video	3.0 Frame Extraction & Spatial Inspection	Sampling Buffer	Representative frame base64 batch for health audit
Level 1 (Sub-Process 4.0)	Clean Payload	4.0 Multimodal LLM Reasoning	Groq Qwen 3.8B Multimodal API	Structured JSON with nutritional, safety, and recipe data
Level 1 (Sub-Process 5.0)	User Metrics	5.0 Metabolic Calculation (Mifflin-St Jeor)	D3: Biometric Target Engine	Calibrated BMR, TDEE, macros, and day-by-day meal schedule